

```
# Cell 1: Database Setup aur Mock Data Insertion
```

```
import sqlite3
```

```
import pandas as pd
```

```
# Database connection
```

```
conn = sqlite3.connect('online_sales.db')
```

```
cursor = conn.cursor()
```

```
# Orders table create karna
```

```
cursor.execute('''
```

```
CREATE TABLE IF NOT EXISTS orders (
```

```
    order_id INTEGER,
```

```
    order_date TEXT,
```

```
    amount REAL,
```

```
    product_id INTEGER
```

```
)
```

```
''')
```

```
cursor.execute("DELETE FROM orders")
```

```
sales_data = [
```

```
    # January 2026
```

```
    (1, '2026-01-05', 1500.00, 501),
```

```
    (2, '2026-01-12', 2500.00, 502),
```

```
    # February 2026
```

```
    (3, '2026-02-03', 4000.00, 501),
```

```
    (4, '2026-02-20', 3500.00, 503),
```

```
    # March 2026
```

```
    (5, '2026-03-15', 6000.00, 504),
```

```
    (6, '2026-03-22', 1200.00, 502),
```

```
    # April 2026
```

```
    (7, '2026-04-10', 8000.00, 505),
```

```
    # May 2026
```

```
    (8, '2026-05-01', 5500.00, 501),
```

```
    (9, '2026-05-18', 4500.00, 504),
```

```
# June 2026
(10, '2026-06-20', 9500.00, 505)
]

cursor.executemany("INSERT INTO orders VALUES (?, ?, ?, ?)", sales_data)
conn.commit()

print("Order Table is Genrated and Data is also loded in this..!")
```

Order Table is Genrated and Data is also loded in this..!

```
# Cell 2: Monthly Revenue aur Order Volume Aggregation Query
trend_query = '''
SELECT
    strftime('%Y', order_date) AS Year,
    strftime('%m', order_date) AS Month,
    COUNT(DISTINCT order_id) AS Order_Volume,
    SUM(amount) AS Total_Revenue
FROM orders
GROUP BY Year, Month
ORDER BY Year ASC, Month ASC;
'''

df_trends = pd.read_sql_query(trend_query, conn)
print("=== MONTHLY SALES TREND RESULTS ===")
df_trends
```

=== MONTHLY SALES TREND RESULTS ===

	Year	Month	Order_Volume	Total_Revenue	
0	2026	01	2	4000.0	
1	2026	02	2	7500.0	
2	2026	03	2	7200.0	
3	2026	04	1	8000.0	
4	2026	05	2	10000.0	
5	2026	06	1	9500.0	

Next steps:

[Generate code with df_trends](#)[New interactive sheet](#)

Cell 3: Trend Line Plot for Monthly Revenue

import matplotlib.pyplot as plt

import seaborn as sns

plt.figure(figsize=(8, 4))

sns.lineplot(data=df_trends, x='Month', y='Total_Revenue', marker='o', color='green', linewidth=2.5)

plt.title('Monthly Revenue Trend (2026)')

plt.xlabel('Month (MM)')

plt.ylabel('Total Revenue (₹)')

plt.grid(True, linestyle='--', alpha=0.6)

plt.show()

